Margin Considerations for Adaptive Radiotherapy (ART)

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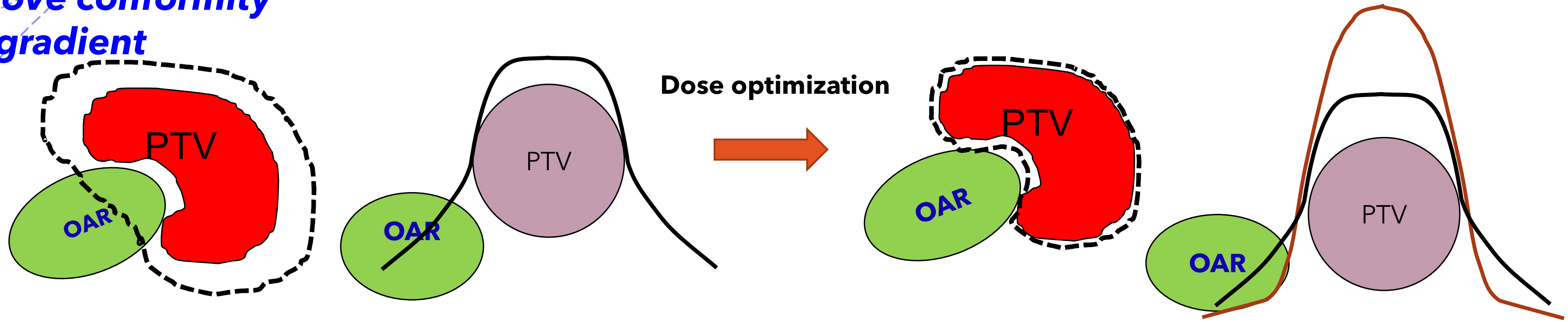
Disclosures

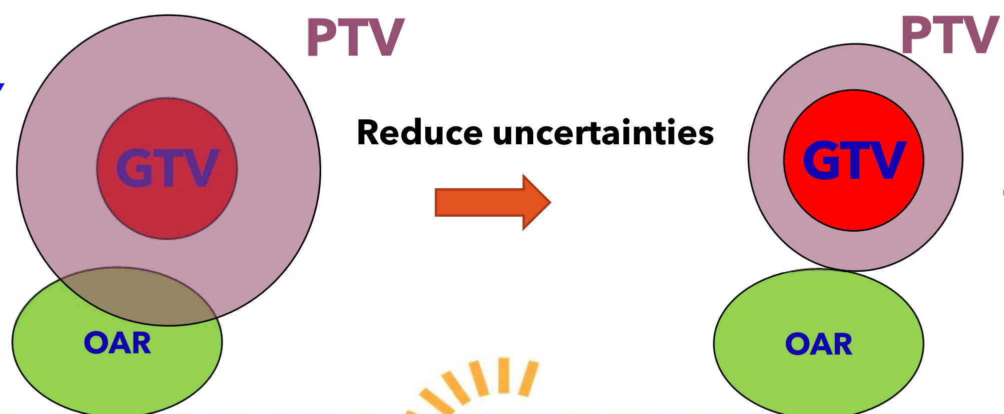
- Consultant for Varian Medical Systems. Medical Dosimetrist Certification Board (MDCB)

Strategies for dose escalation

Improve conformity and gradient

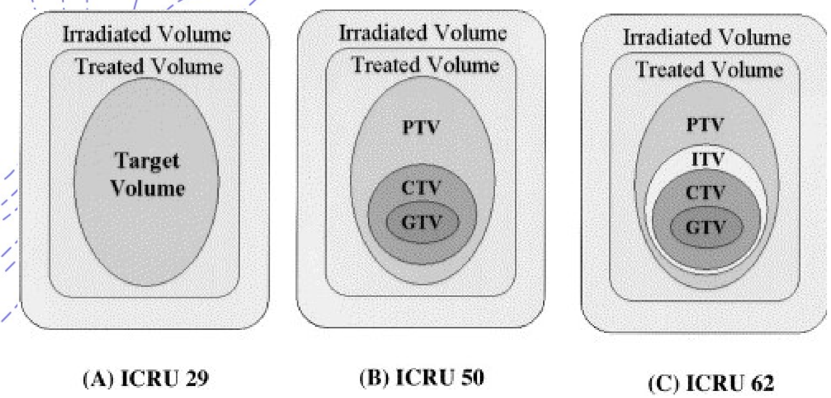


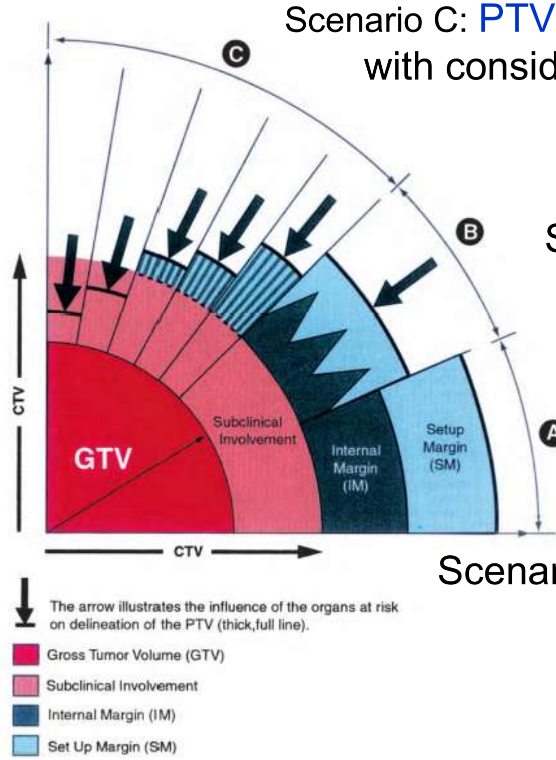
Improve Accuracy



- Advanced imaging and fusion
- Motion management
- Improve machine accuracy
- **Adaptive radiotherapy**
- ….

Target volume definitions – ICRU Reports





Scenario C: PTV = CTV+ global safety margin
with consideration of organs at risk

Scenario B: PTV = CTV+ Margin
Margin is derived by formula
e.g. $\sqrt{\Sigma\sigma^2}$ formalism

Scenario A: PTV = CTV+ IM+ SM

Purdy J. Seminar in Rad Onc. 2004(14): 27-40

ELSEVIER
Seminars in Radiation Oncology
Volume 14, Issue 1, January 2004, Pages 52-64

Errors and margins in radiotherapy

Marcel van Herk [a]

[a] Department of Radiotherapy, The Netherlands Cancer Institute/Antoni van Leeuwenhoek Hospital, Amsterdam, The Netherlands

Available online 27 May 2004.

$$PTV\ Margin = 2.5\Sigma + 0.7\ \sigma$$

Σ : *quadratic sum of SD of all systemic (preparation) errors*

σ: *quadratic sum of SD of all random (execution) errors*

- Margin recipes derived to account for geometric uncertainties
- Uncertainties categorized as **systematic** or **random** errors
- Population-based margin to achieve coverage of the CTV for **p percent** of the patients with the **q%** isodose

p =90% q=95%

Uncertainties to be accounted

- Systematic (preparation): → Influence all fractions
  - Image spatial resolution/contrast
  - Image registration/fusion
  - Contouring intra- and interobserver variability
  - Machine performance uncertainty
  - …

$\mathbf{2.5\Sigma}$

- Random (execution) → Influence each individual fraction
  - Inter-fractional: positioning, deformation, shape variations…
  - Intrafractional motion
  - Machine daily performance uncertainty
  - …

$\mathbf{0.7\,\sigma}$

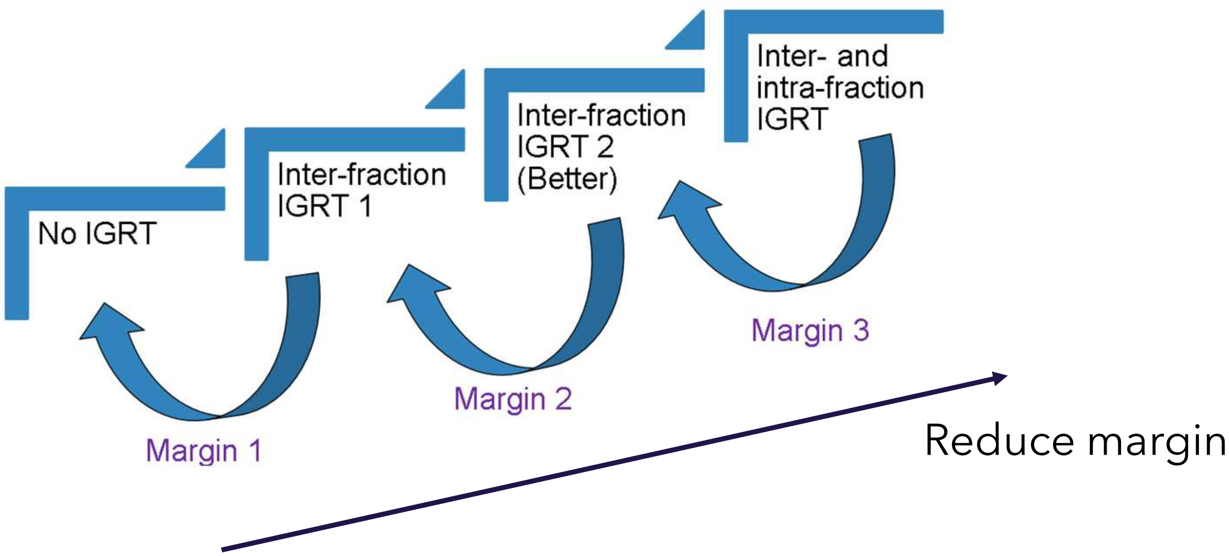
Reduce uncertainty

Reduce margin

$$\boldsymbol{PTV\ Margin = 2.5\Sigma + 0.7\,\sigma}$$

Quantifying uncertainties

- Quantifying uncertainties from multiple sources requires advanced measurement systems



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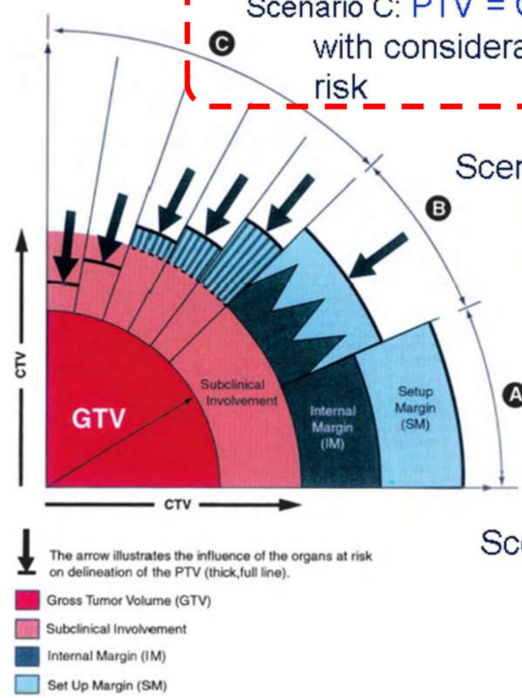
Current clinical practice of margin selection

- Based on published data derived with a better system which can quantify uncertainties, or
- Clinical margins are more often determined as a global safety margin empirically based on institutional and personal experience
- Challenging for margin reduction when new technology introduced

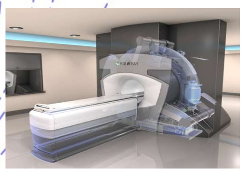
Scenario C: PTV = CTV+ global safety margin
with consideration of the presence of organs at risk

Scenario B: PTV = CTV+ Margin
Margin is derived by formula
e.g. $\sqrt{\Sigma\sigma^2}$ formalism



Scenario A: PTV = CTV+ IM+ SM

Technology → Practice → Clinical outcome



- Superior soft tissue contrast
- Functional and physiological imaging
- Real time dynamic imaging
- No radiation imaging dose

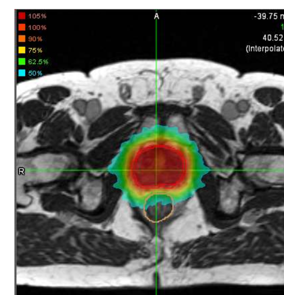
Reduce uncertainties

- Target/OAR delineation
- Treatment setup and verification
- Monitor treatment variations and response
- Motion management

?

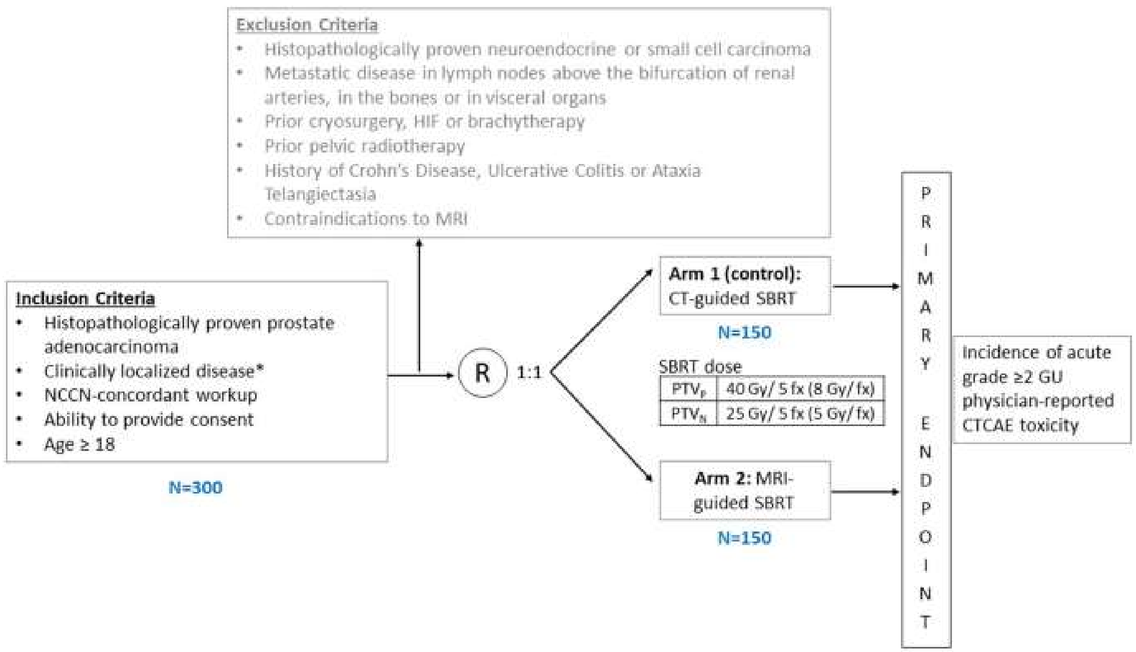
Improve tumor control

Reduce toxicity



A margin reduction study: the MIRAGE trial

Magnetic resonance imaging-guided stereotactic body radiotherapy for prostate cancer (MIRAGE): a phase III randomized trial



Trial design – Key technology differences

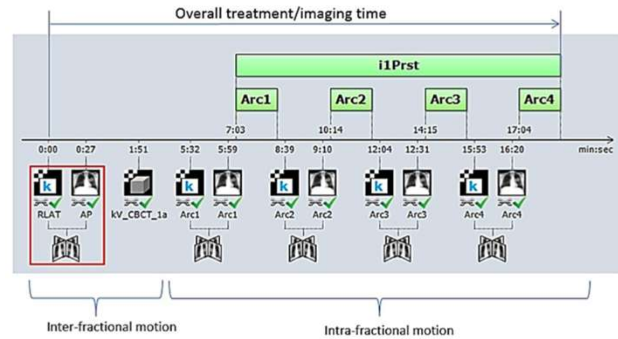
	CBCT-guided arm	MRI-guided arm
Contouring	CT fused with diagnostic MRI	Directly on Sim MRI (no fusion)
Treatment planning	VMAT	IMRT
Patient setup	CBCT (fiducials required)	Daily MRI match
Motion management	None or Repeated x-ray imaging	Real-time cine MRI

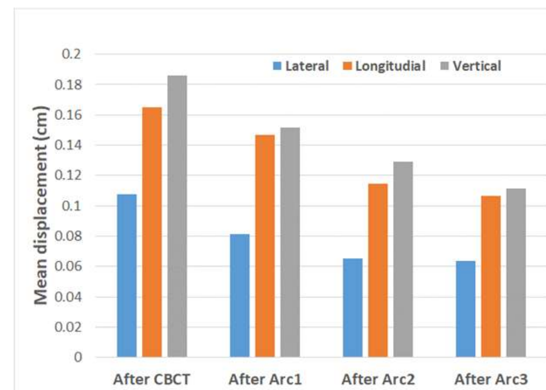
Smaller uncertainties enable smaller PTV margins

Smaller PTV margins reduce toxicity to adjacent OARs

Margins design

- CBCT Arm:
  - Retrospective analysis of 205 prostate SBRT patients (8 Gy × 5 fx) treated with intrafractional imaging
  - Margins calculated for intra-fractional motion : **LR 1.9mm, SI 2.7mm, AP 3.1mm**
  - Accounting for additional uncertainties, we selected an isotropic **4mm margin** for the CT-arm
- MR-Linac Arm: 2mm margin





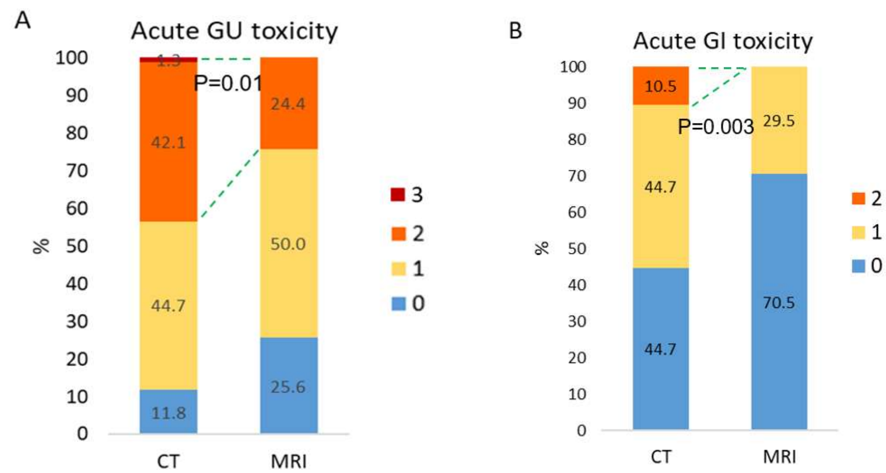
Levin-Epstein R et al. Frontiers in Oncology 2020;10

JAMA Oncology | Original Investigation

Magnetic Resonance Imaging-Guided vs Computed Tomography-Guided Stereotactic Body Radiotherapy for Prostate Cancer
The MIRAGE Randomized Clinical Trial

Amar U. Kishan, MD; Ting Martin Ma, MD, PhD; James M. Lamb, PhD; Maria Casado, BS; Holly Wilhalme, MSc; Daniel A. Low, PhD; Ke Sheng, PhD; Sahil Sharma, BS; Nicholas G. Nickols, MD, PhD; Jonathan Pham, PhD; Yingli Yang, PhD; Yu Gao, PhD; John Neylon, PhD; Vincent Basehart, BS; Minsong Cao, PhD; Michael L. Steinberg, MD

- 156 patients (n=77 CT with 4mm margin and n=79 MRI with 2mm margin)
- Prescription dose 40Gy in 5 fractions to PTV (prostate and proximal SV) ± pelvic nodal RT (25Gy/5fx)
- Primary endpoint: incidence of acute grade ≥2 GU toxicity



MRI arm associated with **a 60% reduction in odds of grade ≥2 GU toxicity** (OR 0.4 [95%CI 0.19-0.84], p=0.02).

Kishan and Ma et al., *JAMA Oncology*. 2023

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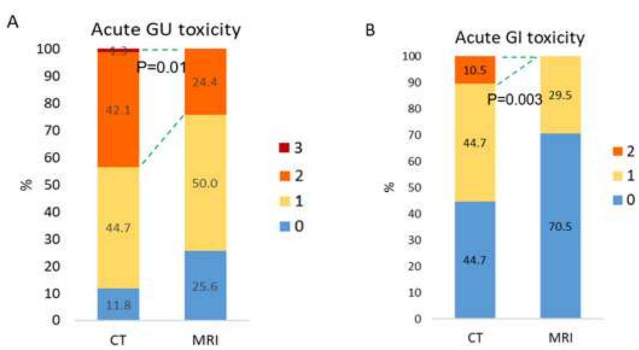
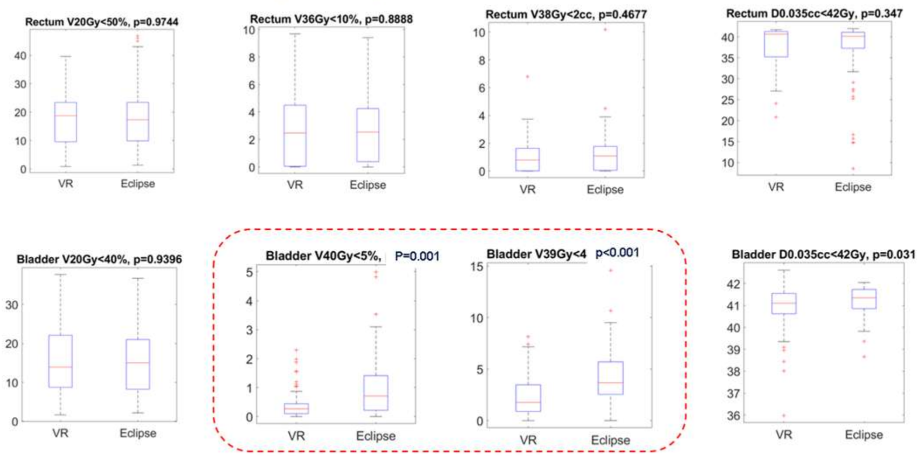
Post-trial analysis

- Multivariable analysis confirmed treatment arm as the strongest predictor of toxicity reduction (over age, IPSS, prostate volume, rectal spacer use, nodal RT, etc.)
- What really contributed to toxicity reduction?
  - Dosimetry
    - Planned dose: margin reduction
    - Delivered dose: setup accuracy and gating
  - Biology (genomics, radiosensitivity)



Planned dosimetry comparison



Dosimetry is not the only contributing factor

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RADIATION ONCOLOGY • BIOLOGY • PHYSICS
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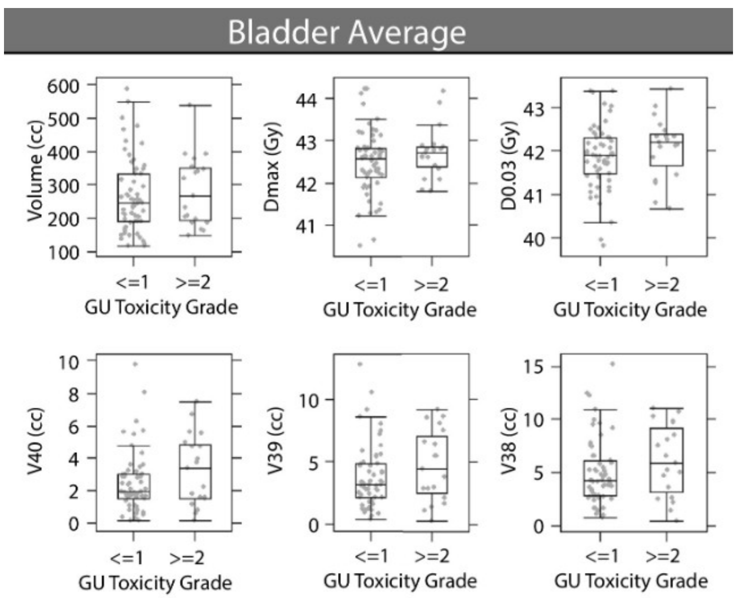
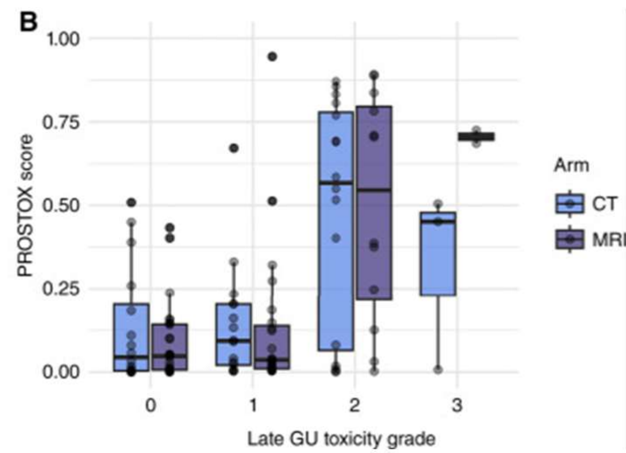
PHYSICS CONTRIBUTION

Impact of Interfractional Bladder and Trigone Displacement and Deformation on Radiation Exposure and Subsequent Acute Genitourinary Toxicity: A Post Hoc Analysis of Patients Treated with Magnetic Resonance Imaging —Guided Prostate Stereotactic Body Radiation Therapy in a Phase 3 Randomized Trial

Dose delivered estimated on setup image



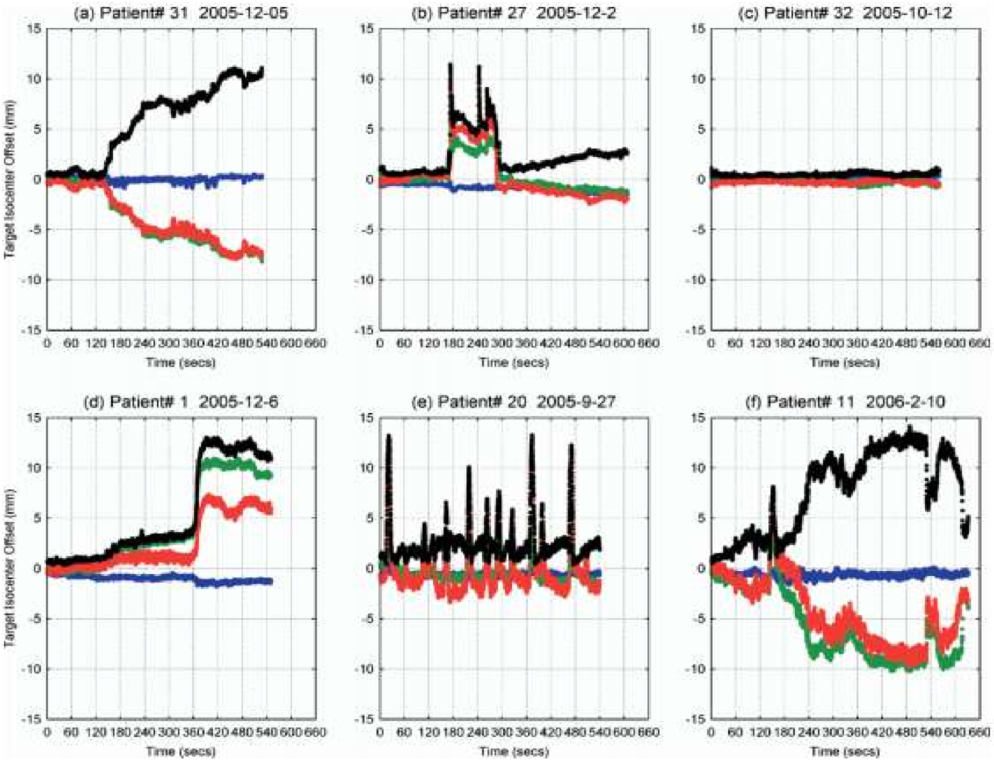


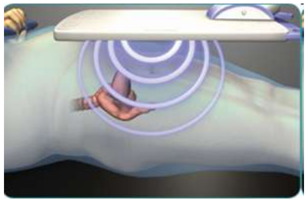
CLINICAL CANCER RESEARCH | TRANSLATIONAL CANCER MECHANISMS AND THERAPY

Validation and Derivation of miRNA-Based Germline Signatures Predicting Radiation Toxicity in Prostate Cancer

Amar U. Kishan[1], Kristen McGreevy[2], Luca Valle[1,3], Michael Steinberg[1], Beth Neilsen[1], Maria Casado[1], Minsong Cao[1], Donatello Telesca[2], and Joanne B. Weidhaas[1]

Intra-fractional prostate motion matters?

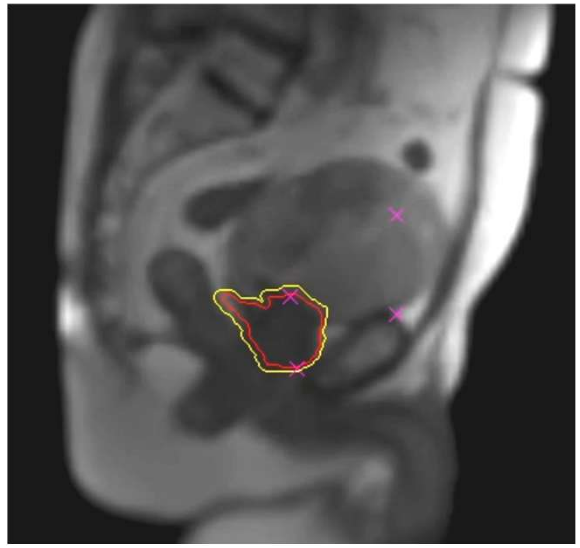


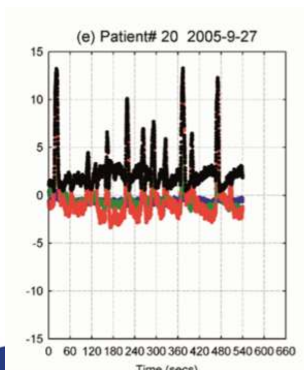


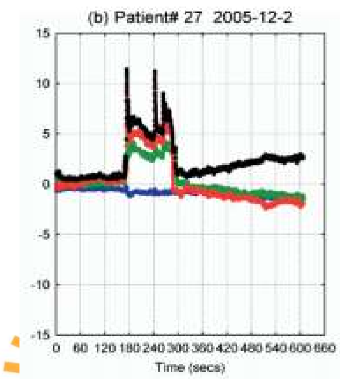
Real-time EM beacon (Calypso) tracking

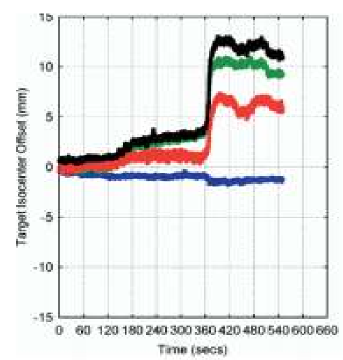
Kupelian, IJROBP, 67: 1088-1098, 2007


R




(e) Patient# 20 2005-9-27
15
10
5
0
-5
-10
-15
0 60 120 180 240 300 360 420 480 540 600 660
Time (secs)


(b) Patient# 27 2005-12-2
15
10
5
0
-5
-10
-15
0 60 120 180 240 300 360 420 480 540 600 660
Time (secs)


Target Isocenter Offset (mm)
15
10
5
0
-5
-10
-15
0 60 120 180 240 300 360 420 480 540 600 660
Time (secs)

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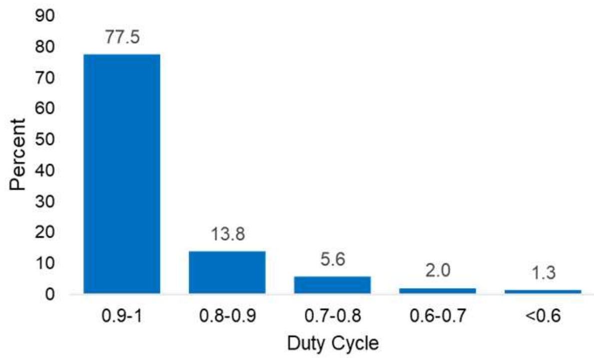
PHYSICS CONTRIBUTION

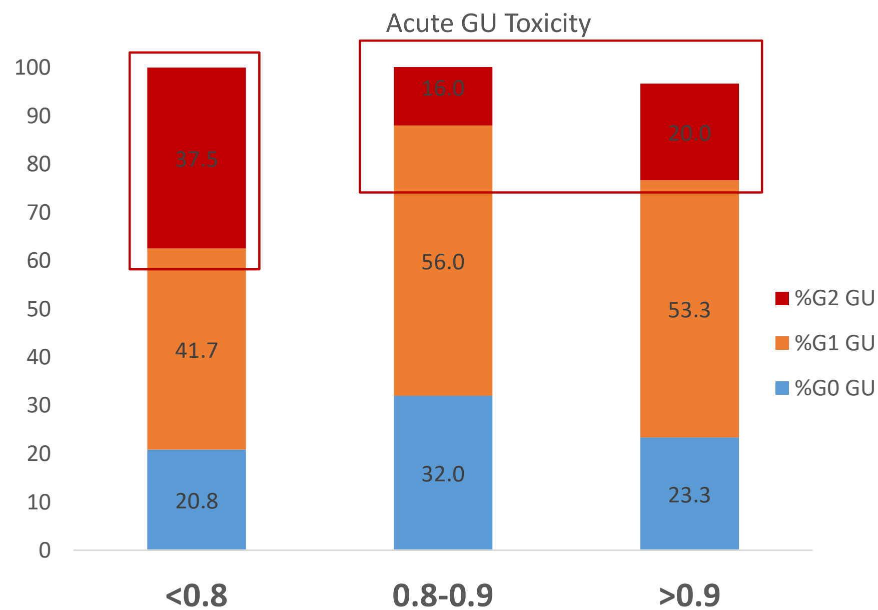
Quantifying Intrafraction Motion and the Impact of Gating for Magnetic Resonance Imaging-Guided Stereotactic Radiation therapy for Prostate Cancer: Analysis of the Magnetic Resonance Imaging Arm From the MIRAGE Phase 3 Randomized Trial



Median duty cycle per fraction: 0.974 (IQR: 0.926 - 0.983)

On a per fraction basis:

- 22.7% had DC<0.9
- 9.0% had DC<0.8



p=0.0647 (2-sided χ^2 test)

Patients with **lower minimum duty cycles** tended to have **higher rates** of acute GU toxicity

Neylon et al. IJROBP 118(5), 2024

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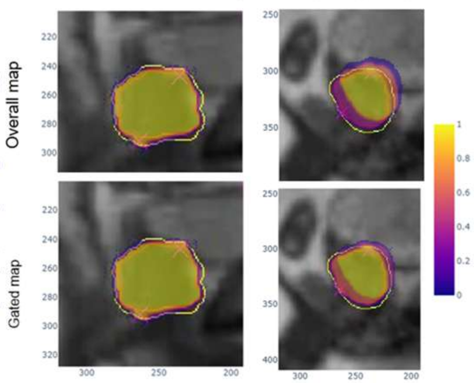
Gating effectively reduced the motion…and toxicity

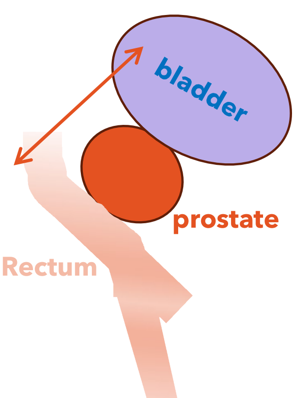
motion-convolved target occupancy map

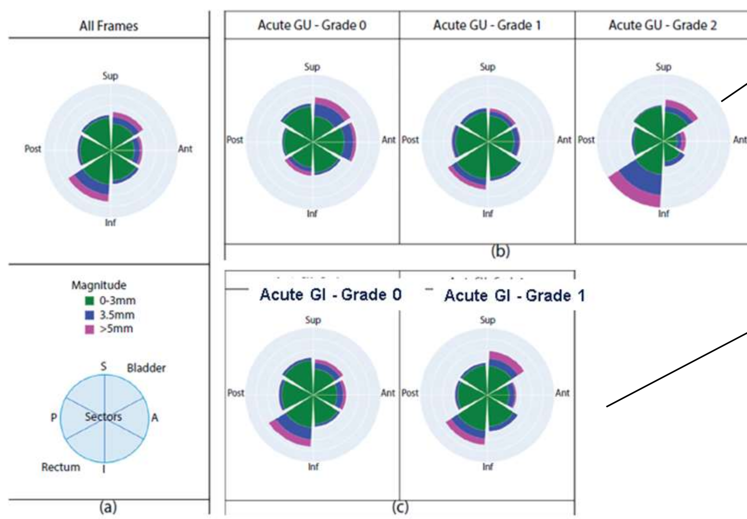


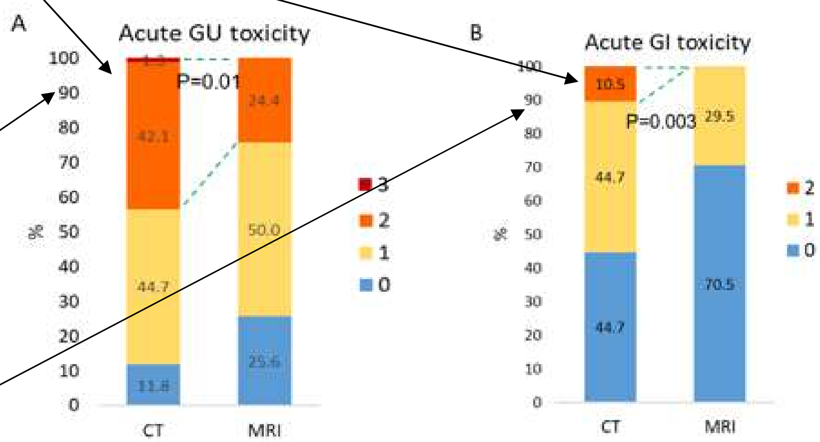
Duty cycle= 0.99 Duty cycle= 0.60

Table 1 Target center of mass motion magnitude statistics

		All frames		Gated frames	
		Mean	IQR	Mean	IQR
All treatments		2.2 mm	1.3-2.8 mm	2.0 mm	1.3-2.4 mm
GDC ≥0.9	84%	1.9 mm	1.2-2.2 mm	1.8 mm	1.2-2.2 mm
GDC <0.9	16%	4.1 mm	3.2-4.7 mm	2.9 mm	2.2-3.8 mm







Neylon et al. IJROBP 118(5), 2024

available at www.sciencedirect.com
journal homepage: www.europeanurology.com

EUROPEAN UROLOGY

European Association of Urology

Brief Report

Magnetic Resonance Imaging Versus Computed Tomography Guidance for Stereotactic Body Radiotherapy in Prostate Cancer: 2-year Outcomes from the MIRAGE Randomized Clinical Trial

Amar U. Kishan [a,b,], James M. Lamb [a], Holly Wilhalme [c], Maria Casado [a], Natalie Chong [a], Lily Zello [a], Jesus E. Juarez [a], Tommy Jiang [a], Beth K. Neilsen [a], Daniel A. Low [a], Yingli Yang [a], John Neylon [a], Vincent Basehart [a], Ting Martin Ma [d], Luca F. Valle [a,e], Minsong Cao [a], Michael L. Steinberg [a]*

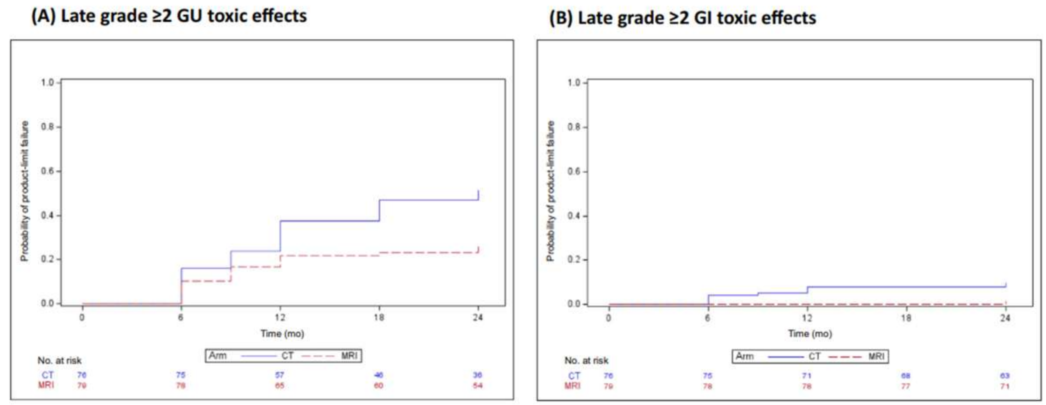


Fig. 1 – Cumulative incidence of late grade ≥2 (A) genitourinary (GU) and (B) gastrointestinal (GI) toxicity, plotted as curves for the probability of product-limit failure for CT-guided and MRI-guided stereotactic body radiotherapy. CT = computed tomography; MRI = magnetic resonance imaging.

Hormone Therapy (Apalutamide) and Image-guided Stereotactic Body Radiation Therapy for the Treatment of Patients With Prostate Cancer, HEATWAVE Trial (HEATWAVE)

ClinicalTrials.gov ID NCT06067269

Sponsor Jonsson Comprehensive Cancer Center

Information provided by Jonsson Comprehensive Cancer Center (Responsible Party)

Last Update Posted 2025-07-15

Image-Guidance and Online Adaptation With Stereotactic Body Radiation Therapy for the Treatment of Localized Prostate Cancer, MANTICORE Trial

ClinicalTrials.gov ID NCT07276438

Sponsor Jonsson Comprehensive Cancer Center

Information provided by Jonsson Comprehensive Cancer Center (Responsible Party)

Last Update Posted 2026-02-05

Margin uncertainties in Adaptive Radiotherapy

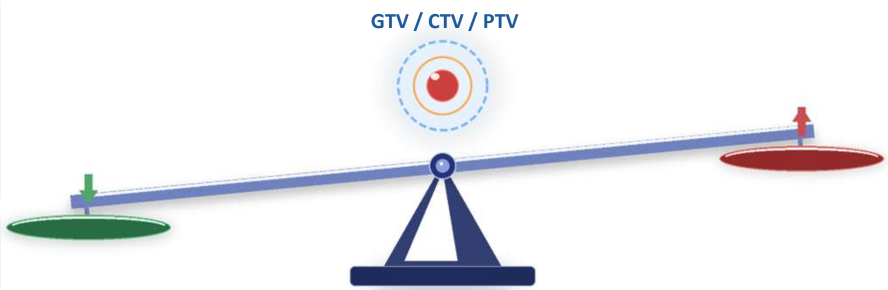
↓ Margin Reduction

Daily recontouring

Converts systematic → random error; reduces its weight in margin formula

Daily replanning

Accounts for tumor regression, progression & organ deformation



Balancing forces that shrink vs. expand the margin

↑ Margin Increase

Prolonged adaptation sessions

Increases intrafractional motion and anatomical drift

Poor imaging / time pressure

Contouring uncertainty rises without high-quality imaging and diagnostic imaging

ART can reduce PTV margins — but only when new uncertainty sources are actively managed

Margin evaluation– Motion during ART

Advances in Radiation Oncology (2024) 9, 101405

advances in radiation oncology

www.advancesradonc.org

Scientific Article

Intrafraction Motion and Margin Assessment for Ethos Online Adaptive Radiotherapy Treatments of the Prostate and Seminal Vesicles

Strahlentherapie und Onkologie (2025) 201:799–806
https://doi.org/10.1007/s00066-024-02346-z

ORIGINAL ARTICLE

Prostate motion in magnetic resonance imaging-guided radiotherapy and its impact on margins

Johannes Kusters[1] · René Monshouwer[1] · Peter Koopmans[1] · Markus Wendling[1] · Ellen Brunenberg[1] · Linda Kerkmeijer[1] · Erik van der Bijl[1]

Workflow A

IGRT workflow

7mm

Workflow C

Adaptive workflow without verification image

5 mm

Workflow B

Adaptive workflow with verification image

3 mm

Adapt to shape only

6 mm

Adapt to shape + Adapt to position

3mm

Addressing intrafractional anatomical drift during adaptive planning is essential

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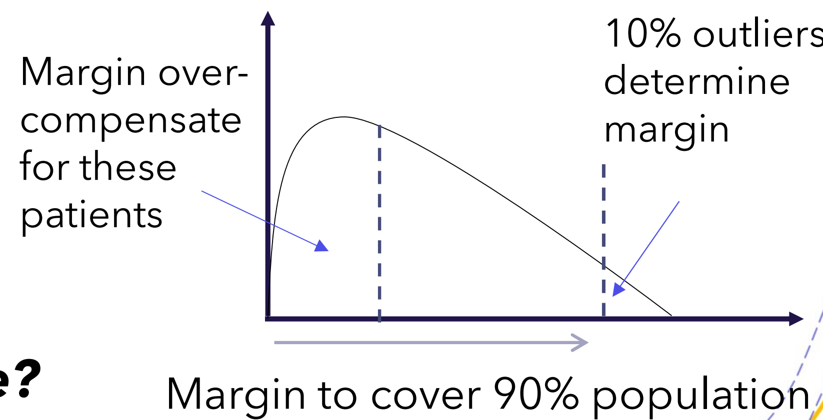
Population-based margin: a good choice for ART?

- Assume geometric errors following a normal distribution
- Prioritize target coverage with insufficient consideration of OAR dose
- **Designed to ensure adequate CTV coverage for a predetermined proportion of patients**
- **Consider uncertainties as static, time-invariant parameters**

$$PTV\ Margin = 2.5\Sigma + 0.7\,\sigma$$

Population-based margin to achieve coverage of the CTV for **90%** of the patients with the **95%** isodose



Can we develop patient-specific margin?

Can we vary margin over treatment course?

Margin in ART – patient-specific and dynamic margins

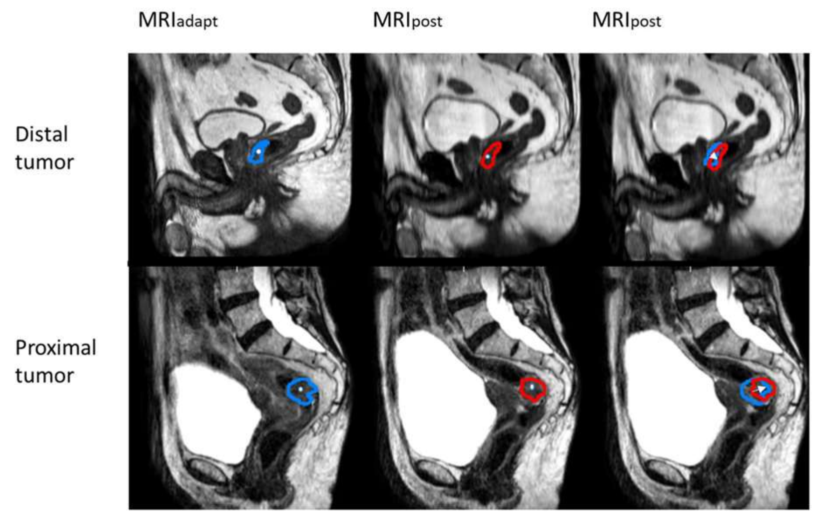
- Patient-specific margins can be determined:
  - Based on patient characteristics (anatomy, physiology and simulation…)
- Margin can be adjusted during treatment course b
  - Based on dose accumulation and motion characteristics from previous treatment fractions
  - Based on specific daily information acquired before treatment delivery

Kensen *et al. Radiation Oncology* (2025) 20:125
https://doi.org/10.1186/s13014-025-02706-8

Radiation Oncology

RESEARCH Open Access

From standardized to individualized margins for online adaptive tumor dose escalation in rectal cancer



Margin individualization based on tumor characteristics (volume, location, circumference)

Table 2 Effect of tumor volume, location and circumference on the magnitude of tumor displacement as estimated by linear mixed effect model

		Left-right		**Anterior-posterior**		**Cranial-Caudal**	
		Estimate	***P*-value**	**Estimate**	***P*-value**	**Estimate**	***P*-value**
Effect of Fraction		0.06	0.332	< 0.01	0.969	0.08	0.217
Effect of Volume		**0.01**	**0.049***	0.01	0.375	< 0.001	0.659
Effect of location in rectum	Distal	0	-	0	-		
	Proximal	-0.04	0.912	**2.15**	**0.001***	**0.97**	**0.043***
Effect of tumor circumference	Semi-circular	0		0			
	Circular	-0.27	0.687	2.27	0.091	0.14	0.888

****statistically significant p-values (p < 0.05)***

Larger margins required for distal tumors in AP (6.5 vs 3.5mm) and CC (6.0 vs 2.0mm) directions than proximal tumors

DOI: 10.1002/acm2.14617

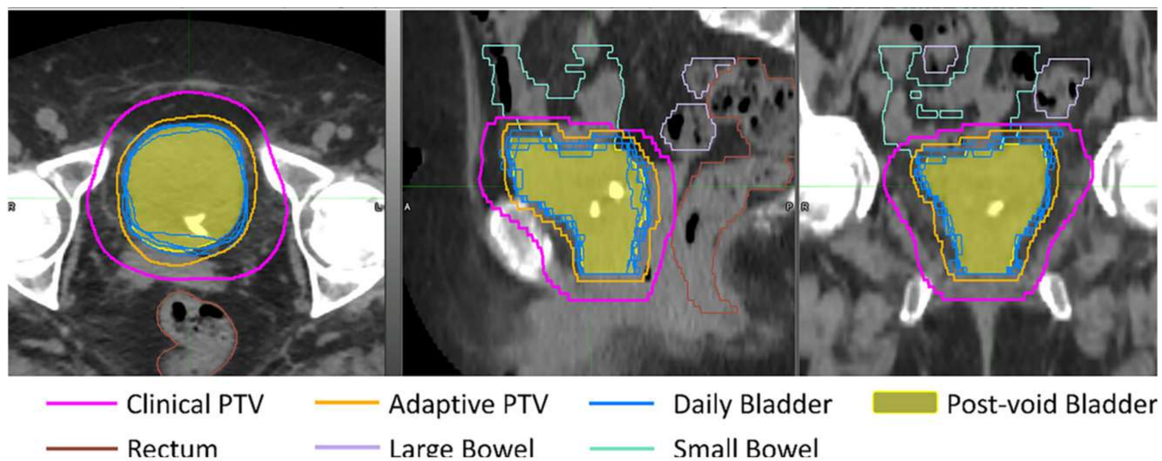
RESEARCH ARTICLE

JOURNAL OF APPLIED CLINICAL MEDICAL PHYSICS

Patient-specific adaptive planning margin for whole bladder radiation therapy

Zhexuan Zhang | Chieh-Wen Liu | Jeremy D. Donaghue | Eric J. Murray | Omar Mian | Ping Xia

- A patient-specific anisotropic adaptive margin recipe for whole bladder irradiation
- Derived from first five fractions of kV-CBCTs; validated and applied to remaining fractions
- Significant PTV volume reduction and superior OAR dose sparing with the adaptive margin



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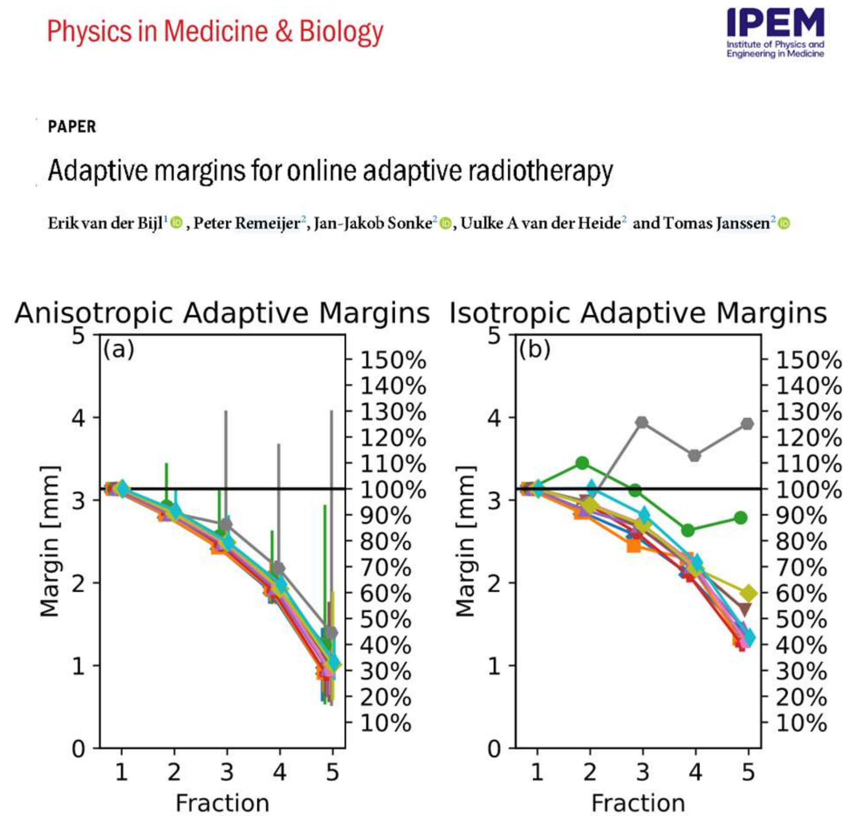
Adaptive margin during Adaptive RT course

- Adaptive margin: the margin for each fraction is determined by accumulated dose from prior fractions
- 1–2 mm margin reduction by eliminating the effective systematic error
- Improved CTV coverage for the 10% of patients under-covered by static population-based margins



Future – do we need PTV margin?

EDITORIAL

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Toward the End of Planning Target Volume Margins With Online Adaptive Radiation Therapy

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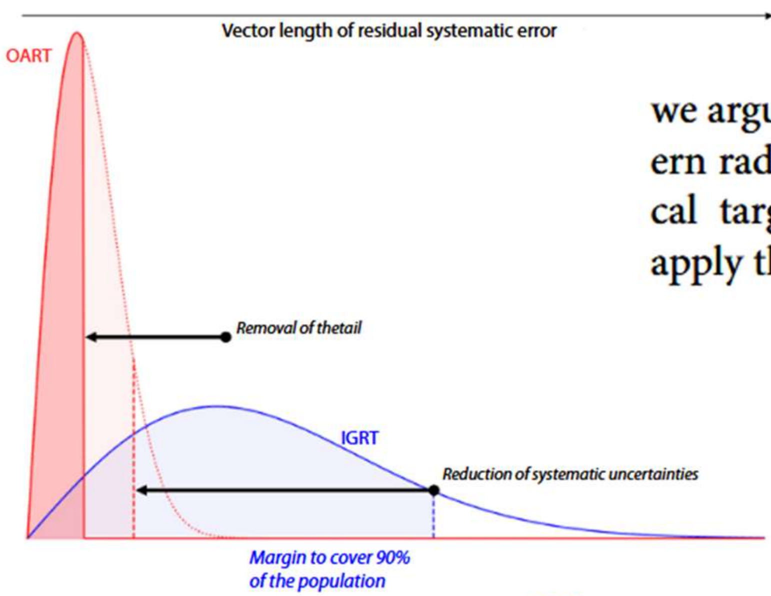

we argue that exploitation of the technical advances in modern radiation therapy makes direct prescription to the clinical target volume (CTV) feasible, removing the need to apply the PTV concept.

Thank You!